Raju Kumar Sah Aerospace Engineer

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Research Profile

Aerospace engineer specializing in experimental and computational rocket propulsion (solid, hybrid and liquid), with hands-on experience in propulsion design, precision fabrication and static-fire testing. Member of the four-person team that built LUME-X1, the first known successful liquid rocket engine built by Nepali students, and led propulsion, structure and fabrication on a solid-fuel rocket flown to 2.6 km; conducted propellant combustion research at IIT Bombay. Research interests: rocket propulsion, high-speed compressible flows, experimental combustion and data-driven CFD. Seeking a funded graduate position to advance propulsion technology through rigorous experiment and simulation.

Honors and Awards

- First known successful liquid rocket engine built by Nepali students: LUME-X1 (2026), covered by Techpana
- Project Trishul (2025): solid-fuel rocket flown to 2,612 m (barometric altitude) against a 3,000 m target
- Winner, PRAKRITI '24 Model Exhibition, Kathmandu University (2024)
- Winner, AIChE K-12 STEM Outreach Competition, AIChE Student Chapter IOE (February 2024)
- Research Support Fund, awarded after these wins
- Mabuchi Motor Award (2023): most compact robot design at ABU Robocon 2023, Phnom Penh, Cambodia

Research Experience

Propellant Research Laboratory, IIT Bombay Sep 2025 – Oct 2025
Research Intern. Supervisor: Prof. Dr. Nagendra Kumar *Mumbai, India*

- **Slurry propellant burn rates:** Prepared HTPB-based slurry propellant (79% AP, 17% HTPB, 3% DOA, 1% charcoal) in a sigma mixer and measured burn rates in a Crawford strand burner over 20–70 bar; fitted Saint-Robert's law by log–log regression. Pressure exponents were 0.41–0.49 across formulations; iron oxide raised the burn-rate coefficient from 1.42 to 1.75 mm/(s·bar^n) versus the no-additive baseline.
- **Gel propellant droplet combustion:** Formulated organic (Thixotrol ST/xylene/ATF) and inorganic (SiO_2/ATF/Al) gels; burned suspended droplets under high-speed imaging and analyzed them with the d^2-law. Non-aluminized organic gels burned steadily ($R^2 > 0.98$); raising silica from 5 to 7.5% increased gel viscosity, and in inorganic gels with 10% silica the best fit was at 25% Al ($R^2 = 0.90$).
- **Laser ignition of AP/HTPB with nano-aluminum:** Ran ignition tests across laser power and ambient pressure, recording piezoelectric, photodiode and pressure-transducer signals. Ignition was inconsistent at 10–20% power and consistent from 40%; piezo ignition delay fell from 46–76 ms at 20% power to 8 ms at 100%.
- **Electrically controlled solid propellant (ECSP):** Crawford-burner tests of a lithium-perchlorate ECSP under varying voltage, pressure and temperature.
- **Hybrid rocket motor:** Supported assembly and static testing of an HTPB/N_2O motor (chamber, nozzle, feed line, sensors) comparing showerhead and swirl injectors; reviewed equilibrium c^*, T_c and I_{sp} versus O/F ratio.
- Sieved ammonium perchlorate into coarse, medium and fine fractions with a tap-sieve shaker.

IIT Bombay Rocket Club (Spaceport America Cup team) Sep 2025 – Oct 2025
Propellant Preparation, Night Shift *Mumbai, India*

- Cast sorbitol/KNO_3 (35%/65%) propellant grains and inspected one by X-ray radiography, which showed slight central density variation and end voids that can cause hot spots and pressure spikes.

SRB Club (Rocketry), Pulchowk Campus Jun 2024 – May 2026
Propulsion, Structure and Fabrication Head *Kathmandu, Nepal*

- Led propulsion and structures for Projects Trishul and LUME-X1, from simulation-driven design to fabrication, integration and testing; contributed to Project Bajra, a senior team's active-fin control rocket.
- Performed thermochemical (NASA CEA) and CFD (ANSYS Fluent) analysis to select O/F ratios and optimize injector flow, chamber thermal loading and nozzle expansion for liquid and solid engines.
- Machined pressure chambers, nozzles, injectors, feed-line fittings, bulkheads and motor mounts on lathes and mills; fabricated glass-fiber composite airframes (body, fins, nosecone) and a lightweight pressure chamber.

Chemical Engineering Laboratory, Pulchowk Campus Jan 2024 – May 2024
Laboratory Research Assistant *Lalitpur, Nepal*

- Tested borax as a flame retardant for raw cotton: untreated cotton ignited at once, while borax-treated cotton resisted ignition under direct flame and formed a protective char layer (qualitative test against an untreated control).

- Studied how borax retards flame: glassy barrier, endothermic dehydration and char promotion.

Selected Projects

LUME-X1: Pressure-Fed Liquid Rocket Engine (Final-Year Thesis, four-person team) 2025 – 2026
Ethanol/GOX. Supervisor: Dr. Sudip Bhattrai

- Led propulsion design, simulation and fabrication; the first hot fire succeeded, and across five static fires measured c^* efficiency was 0.92–0.94 (Tests 4 and 5).

Mach 4.3 Single-Ramp Supersonic Intake Design 2026
Python, ANSYS Fluent

- Designed a shock-on-lip intake using oblique shock theory; automated θ–β–M solutions and throat sizing in Python, giving an analytical total pressure recovery of 0.8874.
- Validated with 2D RANS simulations (k–ω SST): mass flow within 1.03% of the 25 kg/s design value; CFD total pressure recovery 0.8466.

Shock Polar Analysis at Mach 6 (Type I and Type II Interactions) 2026
Python, ANSYS

- Built a Python shock polar solver for two dual-wedge cases: a regular 10°/15° interaction, where the secondary polars intersect, and a 35°/35° Mach-stem case, where they do not; compared with CFD contours.

Project Trishul: Solid-Fuel Rocket 2025
Propulsion, Structure and Fabrication Head

- Flown on 4 January 2025 to 2,612 m (barometric altitude) against a 3,000 m target; second static motor test delivered 3,775 N·s, 91% of the simulated impulse.

Project Bajra: Active-Fin Controlled Rocket (senior team's final-year project) 2025
Contribution: motor casing and nozzle

- Fabricated a lightweight solid rocket motor casing from glass fiber and polyester resin, precision-machined its graphite nozzle and prepared the propellant, meeting the weight and performance limits for flight testing. The control system, flight computer and fin mechanism were developed by the senior team.

ABU Robocon 2023, Nepal Team (Robotics Club) Aug 2022 – May 2026
Design and Fabrication Head, Mabuchi Motor Award *Phnom Penh, Cambodia*

- Designed and machined the robot's mechanical assemblies (base plates, motor mounts, shafts, bearing housings) from CAD to integration; met a late weight limit through weight-relief pocketing and 3D-printed replacements.

Education

Institute of Engineering, Pulchowk Campus, Tribhuvan University 2022 – 2026
B.E. in Aerospace Engineering *Lalitpur, Nepal*

- Aggregate: 70.88%
- Coursework: Compressible Aerodynamics, Advanced Propulsion, Computational Fluid Dynamics, Applied Thermodynamics and Heat Transfer, Fluid Dynamics, Numerical Methods, Control Systems, Aircraft Structures, Flight Dynamics, Aircraft Preliminary Design
- Thesis (final-year project): LUME-X1 liquid rocket engine

Technical Skills

- **Programming/Analysis:** Python (NumPy, SciPy, Pandas, Matplotlib), MATLAB
- **Simulation:** ANSYS Fluent, OpenFOAM, NASA CEA, OpenRocket, OpenMotor, ParaView, XFLR5
- **CAD:** CATIA, Fusion 360
- **Fabrication:** Lathe, milling, welding, composite layup, 3D printing, CNC laser cutting
- **Experimental:** Static-fire testing, propellant formulation, high-speed imaging, piezoelectric/photodiode data acquisition

References

Prof. Dr. Nagendra Kumar, Department of Aerospace Engineering, IIT Bombay (internship supervisor).
Asst. Prof. Dr. Sudip Bhattrai, Pulchowk Campus, Tribhuvan University (thesis supervisor).
Contact details available on request.